A The details of Example 1

To illustrate the proposed verification approach, we analyze the canonical 3-gate circuit shown in Fig. 1a. This circuit implements a CNOT operation between working qubits q_0 and q_1 mediated by an ancilla q_a. The system comprises working qubits $W = \{q_0, q_1\}$ and an ancilla $A = \{q_a\}$. The ancilla is initialized in $|0\rangle_a$ and is expected to return to $|0\rangle_a$.

State Vector Evolution First, we verify the clean ancilla safety by tracing the evolution of computational basis states. Let the input state be $|x, y\rangle_{\mathcal{W}} |0\rangle_a$, where $x, y \in \{0, 1\}$. The evolution of the global state $|\psi\rangle$ proceeds as follows:

$$
\begin{aligned}
|\psi_0\rangle &= |x\rangle_{q_0} |y\rangle_{q_1} |0\rangle_{q_a} \\
&\xrightarrow{\text{CNOT}(q_0,q_a)} |x\rangle_{q_0} |y\rangle_{q_1} |x\rangle_{q_a} \\
&\xrightarrow{\text{CNOT}(q_a,q_1)} |x\rangle_{q_0} |x \oplus y\rangle_{q_1} |x\rangle_{q_a} \\
&\xrightarrow{\text{CNOT}(q_0,q_a)} |x\rangle_{q_0} |x \oplus y\rangle_{q_1} |x \oplus x\rangle_{q_a} = |x\rangle_{q_0} |x \oplus y\rangle_{q_1} |0\rangle_{q_a}
\end{aligned}
$$

The final state is $|\psi_{final}\rangle = |x, x \oplus y\rangle_{\mathcal{W}} \otimes |0\rangle_a$. The ancilla is disentangled and restored to $|0\rangle_a$ for all basis inputs, satisfying the clean ancilla safety definition.

Subspace Invariance We examine the evolution of the linear subspace spanned by valid clean states. Let $\mathcal{V}_{clean} = \text{span}\{|q_0 q_1\rangle |0\rangle_a\}$ be the subspace where the ancilla is in $|0\rangle$. The basis vectors for this subspace are $\{|000\rangle, |010\rangle, |100\rangle, |110\rangle\}$.

1. Detailed Trace of a Representative Basis Vector Let us track the evolution of the basis vector $|100\rangle$ (where $q_0 = 1, q_1 = 0, q_a = 0$). This state is non-trivial because the control qubit q_0 is active.

– **Input State:**
$$|\psi_0\rangle = |1\rangle_{q_0} |0\rangle_{q_1} |0\rangle_{q_a} \in \mathcal{V}_{clean}$$
– **After Gate 1 ($q_0 \to q_a$):** Since $q_0 = 1$, the ancilla flips to $|1\rangle$.
$$|\psi_1\rangle = |1\rangle_{q_0} |0\rangle_{q_1} |\mathbf{1}\rangle_{q_a}$$
Observation: The state vector has left the subspace $\mathcal{V}_{clean}$. It is now in the orthogonal subspace $\mathcal{V}_{dirty}$ (where $q_a = 1$).

– **After Gate 2 ($q_a \to q_1$):** Since $q_a = 1$ (dirty), it activates the second gate, flipping q_1.
$$|\psi_2\rangle = |1\rangle_{q_0} |\mathbf{1}\rangle_{q_1} |1\rangle_{q_a}$$
Observation: The computation is performed correctly, but the state remains outside $\mathcal{V}_{clean}$.

– **After Gate 3 ($q_0 \to q_a$):** Since $q_0 = 1$ still holds, the ancilla flips again ($1 \oplus 1 = 0$).
$$|\psi_{final}\rangle = |1\rangle_{q_0} |1\rangle_{q_1} |\mathbf{0}\rangle_{q_a} \in \mathcal{V}_{clean}$$
Conclusion: The final vector returns to the clean subspace.

2. Summary of Full Basis Evolution Similarly, we can verify the evolution for the remaining basis vectors of $\mathcal{V}_{clean}$. The results are summarized below:

$$\hat{U}|000\rangle = |000\rangle \in \mathcal{V} \qquad \hat{U}|010\rangle = |010\rangle \in \mathcal{V}$$
$$\hat{U}|100\rangle = |110\rangle \in \mathcal{V} \qquad \hat{U}|110\rangle = |100\rangle \in \mathcal{V}$$

Geometric Conclusion Although the intermediate states (like $|\psi_1\rangle$ and $|\psi_2\rangle$) may rotate out of $\mathcal{V}_{clean}$ into the dirty subspace, the unitary U maps every basis vector of $\mathcal{V}_{clean}$ back into $\mathcal{V}_{clean}$. Since the operator is linear, this implies:

$$U(\mathcal{V}_{clean}) = \mathcal{V}_{clean}$$

Thus, the clean ancilla safety is geometrically satisfied.

Commutativity of Reflector Finally, we apply our reduction theorem by verifying the commutativity of the unitary U with the Householder reflection operator $Q_{\mathcal{V}}$. For the clean subspace determined by $|0\rangle_a$, the reflection operator is derived as:

$$Q_{\mathcal{V}} = 2(I_{\mathcal{W}} \otimes |0\rangle\langle 0|_a) - I_{W\cup A} = I_{\mathcal{W}} \otimes Z_a.$$

We verify $[U, Q_{\mathcal{V}}] = 0$ by checking if the operator $Q_{\mathcal{V}}$ remains invariant under the conjugation of each gate in the sequence $U = U_3U_2U_1$. We denote the evolving operator as $Q^{(t)}$.

Initialization: $Q^{(0)} = I_{q_0} \otimes I_{q_1} \otimes Z_{q_a}$.

Step 1: $U_1 = \mathrm{CNOT}(q_0, q_a)$. Using the propagation rule $\mathrm{CNOT}_{c,t}(I_c \otimes Z_t)\mathrm{CNOT}_{c,t} = Z_c \otimes Z_t$:

$$Q^{(1)} = U_1 Q^{(0)} U_1^{\dagger} = Z_{q_0} \otimes I_{q_1} \otimes Z_{q_a}.$$

Step 2: $U_2 = \mathrm{CNOT}(q_a, q_1)$. Here q_a is the control qubit. Since the control qubit's Z-operator commutes with CNOT ($\mathrm{CNOT}_{c,t} Z_c = Z_c \mathrm{CNOT}_{c,t}$):

$$Q^{(2)} = U_2 Q^{(1)} U_2^{\dagger} = Z_{q_0} \otimes I_{q_1} \otimes Z_{q_a}.$$

Step 3: $U_3 = \mathrm{CNOT}(q_0, q_a)$. We apply the conjugation to $Z_{q_0} \otimes Z_{q_a}$. Note that $\mathrm{CNOT}_{c,t}$ maps $Z_c \otimes Z_t$ to $I_c \otimes Z_t$:

$$Q^{(3)} = U_3 Q^{(2)} U_3^{\dagger} = I_{q_0} \otimes I_{q_1} \otimes Z_{q_a}.$$

Conclusion: Since $Q^{(3)} = Q^{(0)} = I_{\mathcal{W}} \otimes Z_a$, we have $UQ_{\mathcal{V}}U^{\dagger} = Q_{\mathcal{V}}$, which implies $[U, Q_{\mathcal{V}}] = 0$. By Theorem 1, the circuit is clean safe.

A.1 Motivating Example: Bridge based GHZ Preparation

To demonstrate the capabilities of our verification tool, we introduce a parameterized family of GHZ state preparation circuits designed for architectures with limited connectivity.

The Bridge Gate Concept In Superconducting Quantum Processors (SQPs), qubit connectivity is typically restricted to a sparse coupling map (e.g., nearest-neighbor). Executing a CNOT between non-adjacent qubits usually requires inserting a chain of SWAP gates, which incurs high overhead (3 CNOTs per SWAP). To mitigate this, Itoko et al. [12] proposed the *Bridge Gate* technique. Instead of swapping states, a bridge gate utilizes an intermediate qubit (ancilla) to mediate the interaction. As shown in Fig. 4, a logical CNOT between control q_c and target q_t is realized via an ancilla a.

Benchmark Design: Cascaded GHZ Generation We construct an n-qubit GHZ state ($|0 \dots 0\rangle + |1 \dots 1\rangle$) by cascading $n-1$ bridge gates. Each step performs a logical CNOT from q_i to q_{i+1} via an ancilla. To evaluate robustness and resource efficiency, we define four variants of this benchmark based on two dimensions: *Safety Property* and *Ancilla Allocation*.

1. Safety Property (Clean vs. Dirty). This dimension defines the assumption on the initial state of the ancilla and the gate construction used:

- **Clean Bridge (3-CNOTs):** As Fig 4 shows, this circuit assumes the ancilla starts in the ground state $|0\rangle$. The sequence is $CX(q_c, a) \rightarrow CX(a, q_t) \rightarrow CX(q_c, a)$. *Verification Challenge:* If the ancilla is initialized to $|0\rangle$, it returns to $|0\rangle$ and performs the correct logic. However, if the ancilla is dirty (e.g., $|1\rangle$), the target qubit suffers a bit-flip error. This requires verifying *Clean Safety*.
- **Dirty Bridge (4-CNOTs):** As Fig 5 shows, this circuit makes no assumption on the ancilla's initial state. It appends a fourth gate, $CX(a, q_t)$, to the Clean Bridge sequence. *Verification Challenge:* This construction guarantees that the ancilla is restored and the logical CNOT is correct regardless of the ancilla's initial state (even if entangled). This requires verifying *Dirty Safety*.

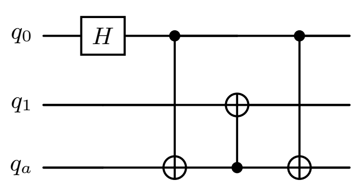

Fig. 4: A Bridge CNOT circuit implementing a mediated operation. The ancilla q_a is used to transfer information and is uncomputed at the end to satisfy dirty safety.

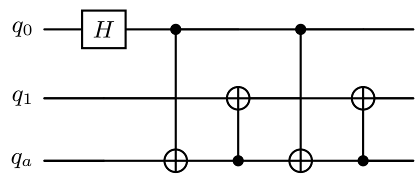

Fig. 5: A Bridge CNOT circuit implemented by a dirty ancilla bewteen q_0 and q_1. The ancilla q_a is used to transfer information and is uncomputed at the end to satisfy dirty safety.

2. Ancilla Allocation (Distinct vs. Shared). This dimension defines how ancilla resources are managed across the cascade:

– **Distinct Ancillae (Linear):** Each logical CNOT uses a fresh, dedicated ancilla. For an n-qubit GHZ state, this requires $n-1$ ancilla qubits. This represents a resource-rich scenario where spatial locality is prioritized. The example circuit is shown is Fig 6.
– **Shared Ancilla (Reused):** A single ancilla is reused sequentially for all $n-1$ logical operations. This represents a resource-constrained scenario. *Verification Challenge:* This introduces complex temporal dependencies. A failure to restore the ancilla in step i will propagate errors to all subsequent steps $j > i$, making verification significantly harder. The example circuit is shown is Fig 7.

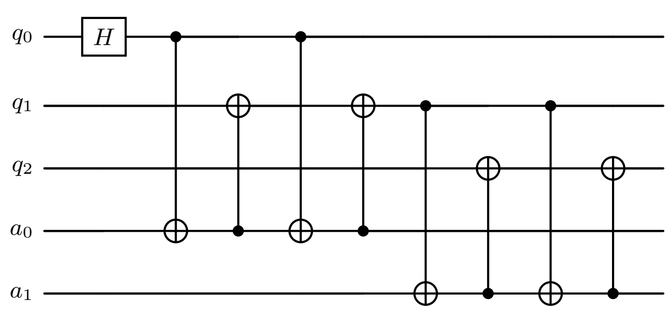

Fig. 6: A cascaded GHZ quantum circuit design utilizing two dirty ancilla qubits (a_0, a_1) to mediate interactions between data qubits.

These variants provide a comprehensive suite for benchmarking quantum verification tools, ranging from simple clean checks to complex, interference-prone dirty safety verification.

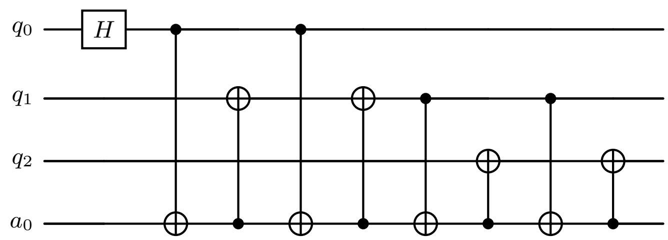

Fig. 7: A cascaded GHZ quantum circuit design utilizing one dirty ancilla qubits a_0 to mediate interactions between data qubits.

B Proof

Proof of Lemma 1. Let $\mathcal{V} = \mathcal{H}_{\mathcal{W}} \otimes \mathrm{span}\{|0\rangle_{\mathcal{A}}\}$.

($\Rightarrow$) Assume U is clean-safe on $\mathcal{A}$. By definition, for any $|\psi\rangle_{\mathcal{W}} \in \mathcal{H}_{\mathcal{W}}$,

$$U(|\psi\rangle_{\mathcal{W}} \otimes |0\rangle_{\mathcal{A}}) = |\psi'\rangle_{\mathcal{W}} \otimes |0\rangle_{\mathcal{A}} \in \mathcal{V}.$$

By linearity, for any $|v\rangle \in \mathcal{V}$, $U|v\rangle \in \mathcal{V}$. Thus,

$$U(\mathcal{V}) \subseteq \mathcal{V}.$$

Since U is unitary, it is an isometry and preserves subspace dimensions:

$$\dim(U(\mathcal{V})) = \dim(\mathcal{V}).$$

For finite-dimensional subspaces, inclusion ($U(\mathcal{V}) \subseteq \mathcal{V}$) and dimension equality imply equality:

$$U(\mathcal{V}) = \mathcal{V}.$$

($\Leftarrow$) Assume $U(\mathcal{V}) = \mathcal{V}$. For any $|\psi\rangle_{\mathcal{W}} \in \mathcal{H}_{\mathcal{W}}$, let $|\phi\rangle = |\psi\rangle_{\mathcal{W}} \otimes |0\rangle_{\mathcal{A}}$. Clearly $|\phi\rangle \in \mathcal{V}$. The assumption implies $U|\phi\rangle \in \mathcal{V}$. By the definition of $\mathcal{V}$, any vector in $\mathcal{V}$ takes the form $|\xi\rangle_{\mathcal{W}} \otimes |0\rangle_{\mathcal{A}}$. Hence, there exists $|\psi'\rangle_{\mathcal{W}} \in \mathcal{H}_{\mathcal{W}}$ such that

$$U(|\psi\rangle_{\mathcal{W}} \otimes |0\rangle_{\mathcal{A}}) = |\psi'\rangle_{\mathcal{W}} \otimes |0\rangle_{\mathcal{A}},$$

which satisfies the definition of clean safety. $\square$

Proof of Lemma 2. Recall that $\mathcal{V} = \mathcal{H}_{\mathcal{W}} \otimes \mathrm{span}\{|0\rangle_{\mathcal{A}}\}$. Let $P_{\mathcal{V}} = I_{\mathcal{W}} \otimes |0\rangle\langle 0|_{\mathcal{A}}$ be the orthogonal projector onto $\mathcal{V}$. The operator $Q_{\mathcal{V}}$ is defined as:

$$Q_{\mathcal{V}} = 2P_{\mathcal{V}} - I.$$

Since the identity operator I commutes with any operator, the commutator relation simplifies to:

$$[U, Q_{\mathcal{V}}] = [U, 2P_{\mathcal{V}} - I] = 2[U, P_{\mathcal{V}}].$$

Thus, $[U, Q_{\mathcal{V}}] = 0 \iff [U, P_{\mathcal{V}}] = 0$. The proof reduces to showing $U(\mathcal{V}) = \mathcal{V} \iff [U, P_{\mathcal{V}}] = 0$.

($\Rightarrow$) Assume $U(\mathcal{V}) = \mathcal{V}$. Consider the operator $R = UP_{\mathcal{V}}U^\dagger$. Since U is unitary, R is an orthogonal projector. For any $|u\rangle \in \mathcal{H}$, let $|v\rangle = P_{\mathcal{V}}(U^\dagger |u\rangle)$. By definition, $|v\rangle \in \mathcal{V}$. Then,

$$R|u\rangle = U|v\rangle \in U(\mathcal{V}) = \mathcal{V}.$$

Thus $\mathrm{Im}(R) = \mathcal{V}$. Since the orthogonal projector onto a subspace is unique,

$$UP_{\mathcal{V}}U^\dagger = P_{\mathcal{V}}.$$

Right-multiplying by U yields $UP_{\mathcal{V}} = P_{\mathcal{V}}U$, i.e., $[U, P_{\mathcal{V}}] = 0$.

($\Leftarrow$) Assume $[U, P_{\mathcal{V}}] = 0$. For any $|v\rangle \in \mathcal{V}$, we have $P_{\mathcal{V}}|v\rangle = |v\rangle$. Applying the projection to the transformed vector $U|v\rangle$:

$$P_{\mathcal{V}}(U|v\rangle) = U(P_{\mathcal{V}}|v\rangle) = U|v\rangle.$$

Since $P_{\mathcal{V}}$ acts as the identity only on $\mathcal{V}$, this implies $U|v\rangle \in \mathcal{V}$. Thus, $U(\mathcal{V}) \subseteq \mathcal{V}$. Since U is an isometry on a finite-dimensional space, inclusion implies equality:

$$U(\mathcal{V}) = \mathcal{V}.$$ □

Proof of Lemma 3. ($\Rightarrow$) Assume U is dirty safe with respect to the full ancilla register $\mathcal{A} = \{a_1, \ldots, a_m\}$. By definition, there exists a unitary V on $\mathcal{H}_{\mathcal{W}}$ such that $U = V \otimes I_{\mathcal{A}}$, which can be expanded as:

$$U = V \otimes I_{a_1} \otimes \cdots \otimes I_{a_m}.$$

Then, for any $j \in \{1, \ldots, m\}$, any state $|\varphi\rangle_{a_j} \in \mathcal{H}_{a_j}$, and any state $|\psi\rangle_{\mathcal{W}\cup(\mathcal{A}\setminus\{a_j\})} \in \mathcal{H}_{\mathcal{W}\cup(\mathcal{A}\setminus\{a_j\})}$, we have

$$U(|\psi\rangle_{\mathcal{W}\cup(\mathcal{A}\setminus\{a_j\})} \otimes |\varphi\rangle_{a_j}) = (V \otimes I_{\mathcal{A}\setminus\{a_j\}})\,|\psi\rangle_{\mathcal{W}\cup(\mathcal{A}\setminus\{a_j\})} \otimes |\varphi\rangle_{a_j},$$

which shows U is dirty safe with respect to a_j alone.

($\Leftarrow$) Assume for every $a_j \in \mathcal{A}$, U is dirty safe on a_j. That is,

$$\forall j \in \{1, \ldots, m\},\ \exists V_j \quad \text{on} \quad \mathcal{H}_{\mathcal{W}\cup(\mathcal{A}\setminus\{a_j\})},$$

such that

$$U = V_j \otimes I_{a_j}.$$

Since this holds for all $j = 1, \ldots, m$, U acts as the identity on every ancilla qubit independently. Hence, U acts as the identity on the entire ancilla register $\mathcal{A}$:

$$U = V \otimes I_{\mathcal{A}}$$

for some unitary V on $\mathcal{H}_{\mathcal{W}}$. □

Proof of Theorem 1. This result follows directly from the chain of equivalences established in Lemma 1 and Lemma 2.

First, by Lemma 1, clean safety is equivalent to the invariance of the clean subspace $\mathcal{V}$:

$$U \text{ is clean-safe on } \mathcal{A} \iff U(\mathcal{V}) = \mathcal{V}.$$

Second, by Lemma 2, the invariance of subspace $\mathcal{V}$ is equivalent to the commutation of U with the operator $Q_{\mathcal{V}}$:

$$U(\mathcal{V}) = \mathcal{V} \iff [U, Q_{\mathcal{V}}] = 0.$$

Combining these two equivalences yields:

$$U \text{ is clean-safe on } \mathcal{A} \iff [U, Q_{\mathcal{V}}] = 0. \qquad \square$$

Proof of Theorem 2. Let $\mathcal{V}_\varphi = \mathcal{H}_{\mathcal{W}} \otimes \text{span}\{|\varphi\rangle_{\mathcal{A}}\}$ and $P_{\mathcal{V}_\varphi} = I_{\mathcal{W}} \otimes |\varphi\rangle\langle\varphi|_{\mathcal{A}}$. The proof proceeds by establishing a chain of equivalences.

1. Clean safety $\iff$ Subspace invariance

- By Definition, U is clean-safe for $|\varphi\rangle_{\mathcal{A}}$ implies that for all $|v\rangle \in \mathcal{V}_\varphi$, $U|v\rangle \in \mathcal{V}_\varphi$. Thus, $U(\mathcal{V}_\varphi) \subseteq \mathcal{V}_\varphi$.
- Since U is unitary (isometry) and $\dim(\mathcal{V}_\varphi) < \infty$:

$$U(\mathcal{V}_\varphi) \subseteq \mathcal{V}_\varphi \implies U(\mathcal{V}_\varphi) = \mathcal{V}_\varphi.$$

2. Subspace invariance $\iff$ Commutativity

- Assume $U(\mathcal{V}_\varphi) = \mathcal{V}_\varphi$. Then the projection onto the subspace is invariant under conjugation by U:

$$UP_{\mathcal{V}_\varphi}U^\dagger = P_{\mathcal{V}_\varphi} \iff UP_{\mathcal{V}_\varphi} = P_{\mathcal{V}_\varphi}U \iff [U, P_{\mathcal{V}_\varphi}] = 0.$$

- Since $Q_{\mathcal{V}_\varphi} = 2P_{\mathcal{V}_\varphi} - I$, and $[U, I] = 0$:

$$[U, P_{\mathcal{V}_\varphi}] = 0 \iff [U, 2P_{\mathcal{V}_\varphi} - I] = 0 \iff [U, Q_{\mathcal{V}_\varphi}] = 0.$$

Combining **1** and **2**, we conclude:

$$U \text{ is clean-safe for } |\varphi\rangle_{\mathcal{A}} \iff [U, Q_{\mathcal{V}_\varphi}] = 0. \qquad \square$$

Proof of Theorem 3. ($\Rightarrow$) Assume U is dirty-safe on $\mathcal{A}$. By Lemma 3, U acts as the identity on the ancilla register $\mathcal{A}$ (up to a unitary on $\mathcal{W}$). That is, $U = V_{\mathcal{W}} \otimes I_{\mathcal{A}}$. Since the identity operator $I_{\mathcal{A}}$ commutes with any operator on $\mathcal{H}_{\mathcal{A}}$, U commutes with $Q_{(a_i,\varphi)}$ for any a_i and any φ.

($\Leftarrow$) Assume the commutativity condition holds:

$$\forall\, a_i \in \mathcal{A},\ \forall\, |\varphi\rangle \in \{|\varphi_1\rangle, |\varphi_2\rangle\}: \quad [U,\, Q_{(a_i,|\varphi\rangle)}] = 0.$$

By Lemma 3, it suffices to prove that U acts as the identity on each individual ancilla qubit a_i (i.e., U is dirty-safe on each a_i).

Fix an arbitrary ancilla qubit a_i. By Theorem 2, the commutativity conditions imply that U is clean-safe for both $|\varphi_1\rangle_{a_i}$ and $|\varphi_2\rangle_{a_i}$. Let us decompose the action of U relative to the basis aligned with $|\varphi_1\rangle$. Without loss of generality, identify $|\varphi_1\rangle$ with the basis state $|0\rangle$ and let $|0\rangle^\perp = |1\rangle$. Since U is clean-safe for $|0\rangle$, it must be block-diagonal with respect to this basis:

$$U = V_0 \otimes |0\rangle\langle 0|_{a_i} + V_1 \otimes |1\rangle\langle 1|_{a_i} \,,$$

where V_0, V_1 are unitaries on the remaining qubits.

Now consider the second state $|\varphi_2\rangle$. Since $0 < |\langle\varphi_1|\varphi_2\rangle| < 0$, we can write $|\varphi_2\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ with non-zero coefficients $\alpha, \beta \in \mathbb{C} \setminus \{0\}$. Applying U to an arbitrary state $|\psi\rangle$ on the working register tensored with $|\varphi_2\rangle$:

$$U(|\psi\rangle \otimes |\varphi_2\rangle) = \alpha(V_0\,|\psi\rangle) \otimes |0\rangle + \beta(V_1\,|\psi\rangle) \otimes |1\rangle \,.$$

Since U is also clean-safe for $|\varphi_2\rangle$, the output must separate as $|\psi'\rangle \otimes |\varphi_2\rangle = |\psi'\rangle \otimes (\alpha\,|0\rangle + \beta\,|1\rangle)$. Matching the coefficients for $|0\rangle$ and $|1\rangle$ implies:

$$V_0\,|\psi\rangle = |\psi'\rangle \quad \text{and} \quad V_1\,|\psi\rangle = |\psi'\rangle \,.$$

Thus $V_0 = V_1$. Consequently, $U = V_0 \otimes (|0\rangle\langle 0| + |1\rangle\langle 1|) = V_0 \otimes I_{a_i}$. This proves that U acts trivially on the ancilla qubit a_i. Since this holds for all $a_i \in A$, U is dirty-safe on A. □

C Implementation Details of Repair Components

C.1 Entanglement Check via Partial Trace

To efficiently verify this, we reduce the problem to a purity computation defined by a partial trace followed by a trace calculation. Specifically, we calculate the reduced operator $\mathrm{O}_{a,\varphi} = \mathrm{Tr}_{\mathcal{W}\cup\mathcal{A}/\{a\}}(Q_{(a,|\varphi\rangle)})$ and check if $\mathrm{Tr}(\mathrm{O}^2_{a,\varphi}) = 1$ for $\varphi \in \{0, +\}$. A result of $\mathrm{Tr}(\mathrm{O}^2_{a,\varphi}) < 1$ for any $\varphi \in \{0, +\}$, indicates that the circuit causes entanglement between the ancilla a and other qubit, otherwise the error is separable with $\mathrm{Tr}(\mathrm{O}^2_{a,\varphi}) = 1$.

C.2 Angle Calculation

Recall $Q_{(a,|0\rangle)} = I_{\mathcal{W}\cup\mathcal{A}/\{a\}} \otimes Z$ and $Q_{(a,|+\rangle)} = I_{\mathcal{W}\cup\mathcal{A}/\{a\}} \otimes X$. $Q'_{(a,|0\rangle)} = UQ_{(a,|0\rangle)}U^\dagger$ and $Q'_{(a,|+\rangle)} = UQ_{(a,|+\rangle)}U^\dagger$. Since the error is separable, Then there exists $O_{a,0}$ and $O_{a,+}$ that are both unitary, satisfying

$$Q'_{(a,|0\rangle)} = I_{\mathcal{W}\cup\mathcal{A}/\{a\}} \otimes O_{a,0}$$

and

$$Q'_{(a,|+\rangle)} = I_{\mathcal{W}\cup\mathcal{A}/\{a\}} \otimes O_{a,+}.$$

Both error occurs If `LogicError` and `PhaseError` both occur, then we find θ_{z1}, θ_x, θ_{z2} which satisfy:

$$R_Z(\theta_{z1})R_X(\theta_x)R_Z(\theta_{z2}) \cdot O_{a,0} \cdot R_Z(\theta_{z1})R_X(\theta_x)R_Z(\theta_{z2})^\dagger = Z$$

and:

$$R_Z(\theta_{z1})R_X(\theta_x)R_Z(\theta_{z2}) \cdot O_{a,+} \cdot R_Z(\theta_{z1})R_X(\theta_x)R_Z(\theta_{z2})^\dagger = X.$$

Existence and Uniqueness of the Solution. The problem of finding the repair angles $(\theta_{z1}, \theta_x, \theta_{z2})$ is equivalent to constructing the inverse unitary $U_{\mathrm{err}}^\dagger$ such that the corrected state aligns with the computational basis. **Existence** is guaranteed by the Euler decomposition theorem, which states that any element of the special unitary group SU(2) can be parameterized by three real rotations $R_Z(\gamma)R_X(\beta)R_Z(\alpha)$. Since the accumulated error U_{err} is a single-qubit unitary operator, there strictly exists a set of angles that satisfies the alignment constraints for both the $|0\rangle$ and $|+\rangle$ states simultaneously.

Regarding **Uniqueness**, the standard Euler decomposition is generally unique up to coordinate singularities (Gimbal lock) and periodicities. However, we enforce a strict unique solution within the principal interval $(-\pi, \pi]$ through our constructive algorithm. First, the periodicity ambiguity is resolved by the use of the two-argument arctangent function (arctan2), which maps all inputs deterministically to the principal range. Second, the potential degeneracy at the poles (where $\theta_x = 0$ or π, causing θ_{z1} and θ_{z2} to become linearly dependent) is resolved by the sequential nature of our derivation. Specifically, θ_{z1} is uniquely determined by the projection of the Z-vector onto the Y-Z plane; once fixed, θ_x and θ_{z2} are subsequently determined by the remaining deviations. Thus, for any physically valid error unitary, our analytic procedure yields exactly one repair sequence.

`LogicError` only or `PhaseError` only When `LogicError` is detected only, then we find $\theta \in (-\pi, \pi]$ satisfying:

$$R_X(\theta)O_{a,0}R_X(\theta)^\dagger = Z.$$

When `PhaseError` is detected only, then we find $\theta \in (-\pi, \pi]$ satisfying:

$$R_Z(\theta)O_{a,+}R_Z(\theta)^\dagger = X.$$

The existence and uniqueness of the solution θ are guaranteed by the algebraic constraints imposed by the diagnosis phase. For a pure `LogicError`, the commutativity check $[U_{\mathrm{err}}, X] = 0$ restricts the error unitary to the subgroup generated by the Pauli-X operator. Geometrically, this confines the Bloch vector of the ancilla to the Y-Z plane. Consequently, the optimization problem reduces to finding a unique rotation angle on a unit circle that aligns the vector with the north pole $|0\rangle$, which is analytically solvable via $\theta = \arctan 2(r_y, r_z)$. Similarly, a pure `PhaseError` implies $[U_{\mathrm{err}}, Z] = 0$, constraining the state to the equatorial

X-Y plane. The solution is thus the unique azimuthal angle required to restore the $|+\rangle$ state, derived as $\theta = \arctan 2(-r_y, r_x)$. In both cases, the dimensionality reduction from the full Bloch sphere to a single great circle ensures a one-to-one mapping between the error syndrome and the repair angle within the principal branch.

D More Experiment Results about Soundness and Scalability of Verification

Table 3 shows more detail comparing performance of different backends.

Table 3: **Comparison of verification efficiency and results.** Time is in seconds. "–" indicates the step is skipped due to timeout/memout.

Benchmark	N	Depth	Matrix Time	Matrix Output	Quokka-Sharp Time	Quokka-Sharp Output	QCEC Time	QCEC Output
Grover$_F$	7	55	0.1	True	6.2	True	0.2	True
	9	99	1.0	True	Timeout	—	0.3	True
	11	195	38.3	True	Timeout	—	0.3	True
	19	1603	Memout	—	Timeout	—	0.4	True
	29	14771	Memout	—	Timeout	—	1.9	True
	35	51459	Memout	—	Timeout	—	8.0	True
	39	115779	Memout	—	Timeout	—	25.9	True
Grover$_{r1}$	99	387	Memout	—	7.0	True	1.0	True
	179	707	Memout	—	21.8	True	2.0	True
	239	947	Memout	—	41.9	True	3.1	True
	319	1267	Memout	—	68.3	True	5.4	True
	399	1587	Memout	—	112.2	True	6.6	True
	479	1907	Memout	—	162.4	True	8.6	True
	559	2227	Memout	—	223.8	True	13.3	True
	699	2787	Memout	—	370.0	True	19.5	True
MCX	199	392	Memout	—	0.6	True	2.2	True
	599	1192	Memout	—	3.7	True	8.7	True
	999	1992	Memout	—	10.0	True	18.7	True
	1399	2792	Memout	—	19.2	True	35.3	True
	1599	3192	Memout	—	25.9	True	44.8	True
	1799	3592	Memout	—	31.8	True	57.7	True
	1999	3992	Memout	—	38.9	True	63.1	True
	3999	7992	Memout	—	174.0	True	514.4	True
	5999	11992	Memout	—	414.4	True	Memout	—
	7999	15992	Memout	—	634.4	True	Memout	—
	9999	19992	Memout	—	1115.6	True	Memout	—
Adder	13	44	159.5	True	0.1	True	0.7	True
	599	2388	Memout	—	3.3	True	13.3	True
	999	3988	Memout	—	8.8	True	38.9	True
	1499	5988	Memout	—	19.3	True	93.4	True
	1999	7988	Memout	—	33.1	True	139.3	True
	2999	11988	Memout	—	71.1	True	444.8	True
	3999	15988	Memout	—	141.9	True	1025.3	True
	4999	19988	Memout	—	218.9	True	Memout	—
	5999	23988	Memout	—	317.8	True	Memout	—
Bridge GHZ	1399	2797	Memout	—	2.4	True	27.3	True
	1799	3597	Memout	—	4.5	True	40.8	True
	1999	3997	Memout	—	5.4	True	48.3	True
	2399	4797	Memout	—	7.6	True	74.9	True
	3599	7197	Memout	—	8.0	True	234.2	True
	4399	8797	Memout	—	10.4	True	Memout	—
	4799	9597	Memout	—	11.0	True	Memout	—
Reqomp-MCX	11	9	2.8	PhaseError	0.1	PhaseError	0.3	PhaseError
	13	11	92.4	PhaseError	0.1	PhaseError	0.3	PhaseError
	15	13	Memout	—	0.1	PhaseError	0.3	PhaseError
Identity Random	5	47	0.1	True	0.4	True	0.2	True
	6	53	0.1	True	1.4	True	0.2	True
	7	54	0.2	True	5.1	True	0.2	True
Random Circuit	30	11	—	Memout	0.1	BothError	0.3	BothError
	50	14	—	Memout	0.1	BothError	0.4	BothError
	70	4	—	Memout	0.1	BothError	0.5	BothError